

Fundamental properties

Learning goals

After this video lecture you should...

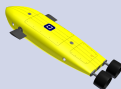
- Be able to validate a mathematical model by ensuring the existence and uniqueness of the solution of an initial value problem

The video lecture is based on

Khalil Chapter 3, Section 3.1

Existence and uniqueness of solutions

Physical system



modelling
→

System model

$$\dot{x} = f(t, x)$$

Existence and uniqueness

Given the system model

$$\dot{x} = f(t, x)$$

and an initial condition

$$x(t_0) = x_0$$

Does there exist a unique solution to the initial value problem (IVP)?

Existence and uniqueness: Sufficient conditions

Theorem 3.1 (Local existence and uniqueness)

If

- $f(t, x)$ is piecewise continuous in t
- $f(t, x)$ satisfies

$$\|f(t, x) - f(t, y)\| \leq L \|x - y\| \quad (\text{Lipschitz condition})$$

$$\forall x, y \in B = \{x \in \mathbb{R}^n \mid \|x - x_0\| \leq r\}$$

$$\forall t \in [t_0, t_1]$$

Then there exists a unique solution of the IVP: $x(t)$ defined on $t \in [t_0, t_0 + \delta]$ for some $\delta > 0$ Local

Existence and uniqueness: Lipschitz

Lipschitz properties

$$\dot{x} = \sqrt[3]{x} \quad x(0) = 0$$

Use Theorem 3.1 to check if this initial value problem has a unique solution.

Existence and uniqueness: Lipschitz

Lipschitz properties

- Locally Lipschitz on $\mathbb{D} \subset \mathbb{R}^n$
- Lipschitz on $\mathbb{D}$
- Globally Lipschitz

Read

Khalil page 89 and Example 3.1

Existence and uniqueness: Lemma 3.1

Lemma 3.1

$$f : [a, b] \times \mathbb{D} \rightarrow \mathbb{R}^n$$

If

$$\left. \begin{array}{l} f(t, x) \text{ continuous in } t \in [a, b] \\ f(t, x) \text{ continuous in } \mathbb{D} \subset \mathbb{R}^n \end{array} \right\} \text{continuous on } [a, b] \times \mathbb{D}$$

$$\frac{\partial f}{\partial x} \text{ exists and is continuous on } [a, b] \times \mathbb{D}.$$

If

$\exists$ convex set $\mathbb{W} \subset \mathbb{D}$ and $\exists$ constant $L \geq 0$ s.t.

$$\left\| \frac{\partial f}{\partial x}(t, x) \right\| \leq L, \quad \forall \quad (t, x) \in [a, b] \times \mathbb{W}$$

Existence and uniqueness: Lemma 3.1 cont.

Lemma 3.1 cont

Then

$$\|f(t,x) - f(t,y)\| \leq L \|x - y\|, \quad \forall t \in [a,b] \\ \forall x,y \in \mathbb{W}$$

i.e. Lipschitz on $\mathbb{W}$.

Existence and uniqueness: Lemma 3.2

Lemma 3.2

$$f : [a, b] \times \mathbb{D} \rightarrow \mathbb{R}^n$$

If

$f(t, x)$ continuous on $[a, b] \times \mathbb{D}$

$\frac{\partial f}{\partial x}$ exists and is continuous on $[a, b] \times \mathbb{D}$

Then f is locally Lipschitz on $\mathbb{D}$.

Read

- Lemma 3.3: Globally Lipschitz
- Theorem 3.2: Global existence and uniqueness
- Example 3.2